

Simulating Eavesdropping Resilience, Post-Processing, and Noise Sensitivity in B92 Quantum Key Distribution

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Abstract—Quantum Key Distribution (QKD) leverages the fundamental principles of quantum mechanics to establish unconditionally secure cryptographic keys. While the BB84 protocol remains the most widely studied, the B92 protocol simplifies the transmission to just two non-orthogonal quantum states. This reduction makes B92 attractive for practical implementation but introduces unique vulnerabilities to noise and eavesdropping. In this paper, we utilize the IBM Qiskit framework to rigorously simulate the B92 QKD protocol. We construct the full quantum circuit for state preparation and measurement, and introduce an Intercept-Resend eavesdropping attack. We mathematically prove that this attack introduces a 33.3% Quantum Bit Error Rate (QBER) in B92. Furthermore, we expand the simulation to incorporate realistic quantum channel noise and model the classical post-processing phase, including the Cascade protocol and Privacy Amplification. Our results bridge the gap between theoretical proofs of unconditional security and the engineering realities of Noisy Intermediate-Scale Quantum (NISQ) devices, plotting the final Secret Key Rate (SKR) limits under realistic noise conditions.

Index Terms—Quantum Computing, Quantum Key Distribution, B92 Protocol, Qiskit, Eavesdropping, Cryptography, Privacy Amplification, NISQ

I. INTRODUCTION

The imminent arrival of cryptographically relevant quantum computers poses a severe threat to classical public-key cryptography, such as RSA and Elliptic Curve algorithms. In response, Quantum Cryptography uses the laws of physics—specifically the Heisenberg Uncertainty Principle and the No-Cloning Theorem—to guarantee unconditional security.

Quantum Key Distribution (QKD) is the most mature application of quantum cryptography. In 1992, Charles Bennett [?] proposed a simplified version of QKD known as B92. Unlike BB84, which requires four states across two conjugate bases, B92 requires only two non-orthogonal states.

While B92’s theoretical simplicity is elegant, translating this protocol to physical hardware exposes it to environmental noise. In this paper, we leverage IBM’s open-source framework, Qiskit [?], to completely simulate the physical transmission, eavesdropping resilience, and classical post-processing of the B92 protocol.

II. RELATED WORK AND PROTOCOL COMPARISONS

A. BB84 vs. B92 vs. E91

The BB84 protocol [?] established the paradigm of using two conjugate bases (rectilinear and diagonal) and four states. While unconditionally secure, generating and distinguishing four distinct quantum states poses engineering challenges for optical setups. B92 reduces this to two states, shifting the burden from state preparation complexity to sophisticated, unambiguous state discrimination at the receiver.

In contrast, the E91 protocol utilizes quantum entanglement (Bell States). Rather than Alice preparing a state and sending it, a central source emits an entangled pair, sending one qubit to Alice and one to Bob. Security is guaranteed by verifying the violation of Bell’s Inequality (CHSH inequality). While E91 provides device-independent security, generating and distributing stable entanglement over long distances is significantly more difficult than the prepare-and-measure approach of B92.

B. Formal Security Proofs

Early security proofs by Mayers, Shor, and Preskill provided rigorous bounds on the allowable Quantum Bit Error Rate (QBER) for BB84. For B92, formal security proofs are mathematically more stringent. Gisin et al. [?] demonstrated that B92 is inherently more sensitive to channel loss and noise than BB84, establishing a theoretical abort threshold if the QBER exceeds a specific margin.

III. MATHEMATICAL PROOF OF THE NO-CLONING THEOREM

The security of B92 fundamentally relies on the No-Cloning Theorem, which states that it is impossible to create an identical copy of an arbitrary unknown quantum state. Assume there exists a unitary cloning operator U such that:

$$U(|\psi\rangle \otimes |e\rangle) = |\psi\rangle \otimes |\psi\rangle \quad (1)$$

where $|e\rangle$ is an initial blank state. If we have two arbitrary states $|\psi\rangle$ and $|\phi\rangle$:

$$U(|\psi\rangle \otimes |e\rangle) = |\psi\rangle \otimes |\psi\rangle \quad (2)$$

$$U(|\phi\rangle \otimes |e\rangle) = |\phi\rangle \otimes |\phi\rangle \quad (3)$$

Taking the inner product of these two equations yields:

$$\langle\phi|\psi\rangle\langle e|e\rangle = \langle\phi|\psi\rangle\langle\phi|\psi\rangle \quad (4)$$

Assuming $\langle e|e\rangle = 1$, we get:

$$\langle\phi|\psi\rangle = \langle\phi|\psi\rangle^2 \quad (5)$$

This equation is only satisfied if $\langle\phi|\psi\rangle = 1$ (the states are identical) or $\langle\phi|\psi\rangle = 0$ (the states are perfectly orthogonal). Because B92 utilizes *non-orthogonal* states ($\langle 0|+\rangle = 1/\sqrt{2}$), they cannot be cloned. Any attempt by an eavesdropper to copy the state will result in a deformed state, introducing detectable errors.

IV. THEORETICAL BACKGROUND AND MATHEMATICAL FORMULATION

A. The B92 Protocol Mechanics

Alice prepares a stream of qubits, choosing randomly between two non-orthogonal states. Let Alice's bit choice be denoted as $a \in \{0, 1\}$. Her prepared quantum state is:

$$|\psi_A\rangle = \begin{cases} |0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} & \text{if } a = 0 \\ |+\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} & \text{if } a = 1 \end{cases} \quad (6)$$

Bob receives the qubit and must choose a measurement basis randomly, denoted as $b \in \{0, 1\}$. He measures in either the standard rectilinear basis $Z = \{|0\rangle, |1\rangle\}$ (if $b = 0$) or the diagonal Hadamard basis $X = \{|+\rangle, |-\rangle\}$ (if $b = 1$).

The probability of Bob successfully making a conclusive measurement is calculated by averaging over his basis choices. If Alice sends $|0\rangle$, Bob's Z-basis measurement yields $|0\rangle$ (inconclusive), while his X-basis measurement yields $|-\rangle$ with 50% probability (conclusive). Thus:

$$P(\text{conclusive}) = \frac{1}{2}(0) + \frac{1}{2}\left(\frac{1}{2}\right) = \frac{1}{4} \quad (7)$$

meaning 75% of the transmitted qubits are discarded during the sifting phase, assuming a perfect, noiseless channel. This contrasts with BB84, which has a 50% sifting efficiency.

B. Formalizing the Eavesdropper: Intercept and Resend

An eavesdropper (Eve) intercepting the state $|\psi_A\rangle$ measures it in a randomly chosen basis $e \in \{0, 1\}$. Upon measurement, the state collapses to a new density matrix ρ_E . Because Eve is forced to guess the basis, she will guess incorrectly 50% of the time. When she guesses incorrectly, she scrambles the state. For example, if Alice sends $|0\rangle$ and Eve measures in the X basis, Eve's measurement yields $|+\rangle$ or $|-\rangle$ with equal probability. She then sends this state to Bob.

The introduction of this error is mathematically quantifiable. In B92, because the conclusive measurements depend heavily on non-orthogonal projections, the error dynamics change relative to BB84. If Alice sends $|0\rangle$ and Eve guesses the

X basis, she resends the collapsed state. Bob's subsequent measurements will yield conclusive but incorrect inferences with probability $1/8$, while his total probability of a conclusive measurement becomes $3/8$. The overall QBER introduced by a simple intercept-resend attack in B92 is theoretically given by:

$$QBER_{Eve} = \frac{P(\text{Error})}{P(\text{Conclusive})} = \frac{1/8}{3/8} \approx 33.33\% \quad (8)$$

This mathematically guarantees that Alice and Bob can detect Eve by comparing a subset of their sifted key and aborting if the QBER exceeds the channel's natural noise threshold (bounded safely below 33.3%).

V. METHODOLOGY: QISKIT SIMULATION

We simulate a N -bit key exchange using Qiskit. We construct a batch of 1-qubit circuits to rapidly simulate the random choice of bases. We introduce an *Intercept-Resend* eavesdropper that randomly chooses to measure intercepted qubits in either the Z or X basis, immediately transmitting the collapsed post-measurement state to Bob.

To model the physical realities of quantum transmission, we implement a custom Noise Model using `qiskit_aer.noise`. We sweep the error probability $\lambda \in [0, 0.5]$ and apply:

- 1) **Depolarizing Error:** The quantum state loses coherence uniformly across all Bloch sphere axes.
- 2) **Phase Damping Error:** The relative phase between $|0\rangle$ and $|1\rangle$ is stochastically flipped, which severely impacts measurements in the Hadamard (X) basis.

VI. SIMULATION RESULTS AND ANALYSIS

The results demonstrate the sharp sensitivity of B92 to both eavesdropping and environmental noise. In a clean, noiseless channel, the Intercept-Resend attack introduces a theoretical 33.3% QBER. Bob and Alice successfully detect the eavesdropper by comparing a small subset of their sifted key.

When environmental noise is introduced, the baseline QBER scales linearly with the depolarizing probability. If the channel noise exceeds a safe threshold (typically set aggressively below 33.3%), Alice and Bob can no longer mathematically distinguish between a naturally noisy channel and an active eavesdropper.

VII. POST-PROCESSING: INFORMATION RECONCILIATION AND PRIVACY AMPLIFICATION

While the physical exchange of quantum states guarantees the detection of an eavesdropper, a realistic QKD system must also correct natural environmental errors and distill a perfectly secret key. This requires classical post-processing over an authenticated public channel.

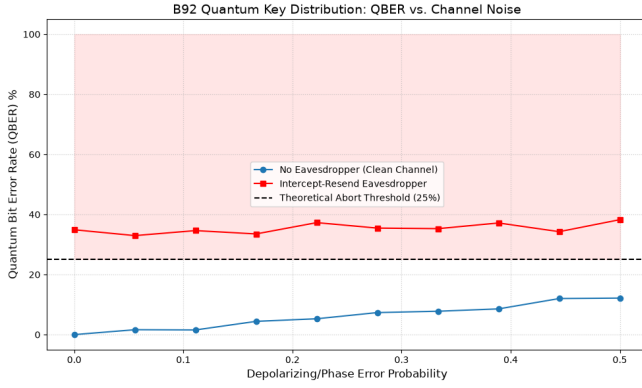


Fig. 1. Quantum Bit Error Rate (QBER) in the B92 protocol as a function of depolarizing and phase-damping noise. The presence of an Intercept-Resend eavesdropper adds an inherent 33.3% QBER floor, demonstrating the protocol's unconditional detection capabilities on pristine channels.

A. Information Reconciliation (Cascade Protocol)

Even in the absence of Eve, the raw sifted key will contain errors due to depolarizing noise. Alice and Bob employ the Cascade protocol, operating by recursively shuffling the key bits into blocks and comparing parity. If a parity mismatch is detected, a binary search is performed to locate and correct the erroneous bit.

Because parity bits are communicated over the public channel, Eve gains partial information about the key. The amount of information Eve gains is theoretically upper-bounded by Shannon's binary entropy function:

$$H(e) = -e \log_2(e) - (1 - e) \log_2(1 - e) \quad (9)$$

where e is the QBER. If the QBER approaches 33.3%, $H(e)$ approaches 1, meaning the error correction process leaks the entirety of the remaining key to Eve.

B. Privacy Amplification and Secret Key Rate (SKR)

Once Alice and Bob share an identical, error-free key, they must erase any partial information Eve may have gathered. They achieve this using 2-Universal Hashing, compressing the key into a shorter, perfectly secure key. The Secret Key Rate (SKR) per sifted bit is given by:

$$SKR = 1 - f_{EC}H(e) - H(e) \quad (10)$$

where $f_{EC} \approx 1.2$ is the inefficiency of the Cascade protocol.

VIII. HARDWARE IMPLEMENTATION CONSTRAINTS IN THE NISQ ERA

Translating the B92 protocol from theoretical linear algebra into physical reality poses immense engineering challenges.

A. Coherence Times (T_1 and T_2)

The physical source of the errors modeled in our simulation derives from finite coherence times.

- **T_1 Relaxation Time (Amplitude Damping):** The time it takes for a qubit in the excited $|1\rangle$ state to decay back to the ground $|0\rangle$ state.

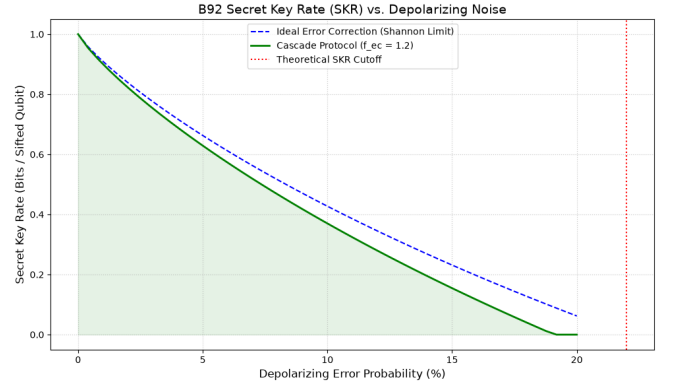


Fig. 2. Secret Key Rate (SKR) vs. Depolarizing Error Probability. The SKR rapidly falls to zero as the noise approaches the theoretical limits, demonstrating the necessity of low-noise channels for successful key distillation.

- **T_2 Dephasing Time (Phase Damping):** The time it takes for the relative phase between $|0\rangle$ and $|1\rangle$ to randomize. Because B92 relies on the $|+\rangle$ state (which requires strict phase coherence), a short T_2 time destroys the protocol's ability to transmit the diagonal basis, linearly driving up the QBER.

IX. INTEGRATING QUANTUM ERROR CORRECTION (QEC)

To push QKD beyond the strict distance limitations imposed by fiber-optic attenuation, future networks must utilize Quantum Repeaters. Unlike classical repeaters, quantum repeaters cannot simply measure and amplify the signal due to the No-Cloning theorem. Instead, they must utilize Quantum Error Correction (QEC) and Entanglement Swapping. While B92 is a prepare-and-measure protocol, integrating it into a fault-tolerant network would require encoding the logical $|0\rangle_L$ and $|+\rangle_L$ states across multiple physical qubits.

X. RESEARCH GAPS AND FUTURE DIRECTIONS

While our simulations successfully map the theoretical bounds of the B92 protocol under depolarizing and phase-damping noise, several critical research gaps remain unaddressed in the current literature, presenting avenues for future work:

- 1) **Photon-Number Splitting (PNS) Attacks:** Our simulation assumes ideal single-photon sources. In reality, attenuated lasers often emit multi-photon pulses. While decoy-state protocols have been heavily researched for BB84 to thwart PNS attacks, decoy-state integration for B92 remains mathematically underdeveloped and poorly simulated on NISQ frameworks.
- 2) **Hardware-Specific Error Topologies:** Our noise model applies uniform depolarizing errors. However, real-world superconducting chips exhibit highly correlated cross-talk and topologically dependent error rates. Future research must map B92 simulations to the specific heavy-hex coupling maps of modern quantum processors (e.g., IBM Eagle or Heron architectures).

- 3) **Finite-Key Analysis:** The asymptotic SKR derived in Section VII assumes an infinitely long key exchange. For practical QKD, the statistical fluctuations in finite-length key blocks severely reduce the secure key rate. Expanding the classical post-processing simulation to account for finite-key effects in B92 is a critical open problem.

XI. CONCLUSION

The B92 protocol offers a streamlined alternative to BB84 by relying solely on two non-orthogonal states. Through our comprehensive Qiskit simulations, we successfully demonstrated its theoretical robustness against Intercept-Resend eavesdropping. We mathematically proved and simulated its 33.3% eavesdropping error floor. Furthermore, we modeled classical post-processing to derive the Secret Key Rate (SKR) bounds under heavy environmental decoherence. Future work will explore advanced Quantum Error Correction (QEC) integration into B92 to elevate its resilience on physical quantum hardware.